

GUILHERME LOPES DIAS

Software Engineer | AI & Automation

Brazil • Remote

Portfolio: guilhermelopesdias.dev

GitHub: github.com/guilhermelopesdias

LinkedIn: linkedin.com/in/guilherme-lobes-dias

PROFESSIONAL SUMMARY

Computer Science student and Software Engineer specializing in Python, AI agent architectures, backend development, and system automation. Experienced in architecting low-latency multimodal voice agents, encrypted real-time communication systems, and hybrid deterministic/LLM classification pipelines. Driven by clean software architecture, algorithmic efficiency, and transparent technical trade-offs.

PROFESSIONAL EXPERIENCE

Software Automation Developer

2025 – Present

Freelance / Independent • Remote (Brazil)

- Architect and deploy end-to-end automation pipelines integrating LLM APIs, custom Python microservices, and n8n platform workflows to eliminate operational manual overhead.
- Build conversational AI agents and autonomous support workflows featuring multi-step state management, API integrations, and robust exception handling.
- Manage end-to-end delivery cycles: problem scoping, workflow modeling, API contract design, and iterative testing.

SELECTED PROJECTS

Merlin — Multimodal Voice AI Agent

Python • LLM APIs • n8n • WebSockets

- Engineered a low-latency, voice-driven AI agent capable of parsing natural-language speech to trigger automated multi-stage system workflows.
- Integrated speech-to-text, LLM reasoning pipelines, and workflow orchestration into a modular pipeline optimized for real-time responsiveness.
- Documented system architecture, event hooks, and computational latency constraints to ensure maintainability and technical clarity.

Nexus — Encrypted Real-Time Communication System

Python • Sockets • AES-128 • Networking

- Architected a client-server messaging platform utilizing raw TCP sockets, custom binary serialization, and end-to-end AES-128 encryption.
- Implemented distributed governance mechanisms and dynamic session state handling across concurrent network client connections.
- Evaluated cryptographic overhead and throughput performance to optimize message latency and payload security.

Giro — Intelligent Transaction Classifier

Python • RegEx • OpenAI API • Data Pipelines

- Developed a full-stack financial automation application featuring a high-speed hybrid transaction classification engine.
- Combined deterministic RegEx pattern matching with LLM fallback queries, drastically reducing manual categorization effort while maintaining high accuracy.
- Built robust data cleaning and normalization pipelines for processing heterogeneous financial records seamlessly.

GraphRoute — Algorithm Benchmarking Engine

Python • Data Structures • Graph Theory

- Implemented and benchmarked fundamental graph traversal algorithms (Dijkstra, A*, BFS, DFS) through an interactive visualization suite.
- Conducted empirical runtime and space complexity analyses across dynamic graph topologies to evaluate real-world performance trade-offs.

TECHNICAL SKILLS

Languages:	Python, TypeScript, SQL, C/C++ (Foundations)
Software Engineering:	Data Structures & Algorithms, Object-Oriented Programming (OOP), REST APIs, Git/GitHub, Linux, Networking (TCP/IP, Sockets)
AI & Automation:	LLM APIs (OpenAI / Gemini), Autonomous AI Agents, Prompt Engineering, Multimodal AI, n8n Workflow Automation
Cybersecurity & Tools:	Ethical Hacking Concepts, NFC/RFID Hardware Protocols (Mifare / ISO 14443-A), Power BI, Regex

EDUCATION & LEADERSHIP

UNIFACS — B.S. in Computer Science

Feira de Santana, BA • Expected Graduation: 2028 (3rd Semester)

Leadership: Class Representative — Acts as official student liaison, communicating academic feedback and coordinating with course faculty.

CERTIFICATIONS & LANGUAGES

- Python Certification
- Santander Academy: Data Science & Analytics
- Cisco Networking Academy: Endpoint Security
- Languages: Portuguese (Native), English (Intermediate / Technical), Spanish (Basic)